$$AM = \frac{\sum fX}{N} = \frac{2357}{140} = 16.84$$

$$GM = Anti\log\left(\frac{\sum f \log X}{N}\right)$$

$$= Anti\log\left(\frac{167.209}{140}\right) = Anti\log(1.1944) = 15.65$$

$$HM = \frac{N}{\Sigma\left(\frac{f}{X}\right)} = \frac{140}{9.6852} = 14.46$$

i.e. 16.84>15.65 >14.46

$\therefore$ $AM > GM > HM$

NOTE

The Harmonic Mean is a measure of central tendency for data expressed as rates such as kilometres per hour, kilometres per litre, periods per semester, tonnes per month etc.

Example 8.14

A's scooter gives an average of 40km a litre while B's scooter gives an average of 30km a litre . Find out the mean, if

(i) each one of them travels 120 km.

(ii) the petrol consumed by both of them is 2 litres per head.

Solution:

(i) Here the distance is constant. Hence harmonic mean is appropriate.

$$HM = \frac{n}{\frac{1}{a}+\frac{1}{b}}$$

$$= \frac{2}{\frac{1}{40}+\frac{1}{30}} = \frac{2}{\frac{7}{120}}$$

$$= \frac{2\times120}{7} = 34.3 \text{ km per litre}$$

(ii) Here the quantity of petrol consumed is fixed i.e 2 litres. Here the arithmetic mean will give the correct answer.

$$\overline{X} = \frac{Total\ distance\ Covered}{Total\ Petrol\ Consumed}$$

$$\overline{X} = \frac{40\times2+30\times2}{4} = 35$$

$\therefore$ Average speed = 35 km per litre.

Example 8.15

A person purchases tomatoes from each of the 4 places at the rate of 1kg., 2kg., 3kg., and 4kg. per rupee respectively .On the average, how many kilograms has he purchased per rupee?

Solution:

Since we are given rate per rupee, harmonic mean will give the correct answer.

$$HM = \frac{n}{\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}}$$

$$= \frac{4}{\frac{1}{1}+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}}$$

$$= \frac{4\times12}{25}$$

= 1.92 kg per rupee.

8.2 Measures of Dispersion

Average gives us an idea on the point of the concentration of the observations about the central part of the distribution. If we know the average alone we cannot form a complete idea about the distribution as will be clear from the following example.

Consider the series:

(i) 7,8,9,10,11 (ii) 3,6,9,12,15 and (iii) 1,5,9, 13,17. In all these cases we see that n, the number of observations is 5 and the mean is 9. If we are given that the mean of 5 observations is 9, we cannot form an idea

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